QUANT

Time, Speed & Distance

Assignment 03

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Time, Speed & Distance Online Assignment -03

1. Three boys, A, B and C start jogging from the same point simultaneously in the same direction at 3 mph, 5 mph and 6 mph respectively on a circular path of diameter 161 yards. After what time will they meet again? (1 mile = 1760 yards)

1. 1035 sec 2. 900 sec 3. 15 min 25 sec 4. 1075 seconds

2. A man takes 20 minutes to row 12 km upstream which is a third more than the time he takes on his way downstream. What is his speed in still water?

1. 41 km/hr 2. 36 km/hr 3. 42 km/hr 4. 45 km/hr

3. In a km race, if A gives B a start of 20 seconds, A wins by 10 meter. Alternatively, if he gives him a 50 meter start, then they finish together. How many seconds start should A give B, so that they finish together?

1. 30 sec 2. 25 sec 3. 27 sec 4. 33.33 sec

4. There is a circular track with a length of 120 metre. Two persons A and B start running together at the same time in the same direction with respective speeds of 40 metre/sec and 25 metre/sec. How many points are there on the circular track where they will meet?

1. 1 2. 2 3. 3 4. 4

5. A train crosses a tree in 20 seconds and a man cycling at 5 mph in the opposite direction in 18 seconds. What is the length of the train?

1. $\frac{1}{4}$ of a mile 2. $\frac{1}{3}$ of a mile 3. $\frac{3}{10}$ of a mile 4. $\frac{2}{5}$ of a mile

6. A boat went down the river for a distance of 40 km. It then turned back and returned to its starting point, having traveled a total of 7 hours. On its retun-trip at a distance of 24 km from the original starting point, it encountered an eraser, which was dropped at the starting point just at the moment at which the boat had started downstream. The downstream speed of the boat is

1. 8 kmph 2. 26 kmph 3. 20 kmph 4. 16 kmph

7. A train takes 39 seconds to completely overtake a cyclist traveling at 5 mph. However, it would have taken 6 seconds lesser to completely cross him if he was traveling in the opposite direction. What is the speed of the train?

1. 122 mph 2. 60 mph 3. 180 mph 4. None of these

8. Prem covers the first half of the distance between Delhi and Chandigarh at 45 km/hr and the rest at 55 km/hr. What is the distance between Delhi and Chandigarh, if he took 4 hours to reach Chandigarh?

1. 200 km 2. 198 km 3. 196 km 4. 49.5 km

9. A and B start running from P to Q and immediately return back to P. When B has covered $3/4^{th}$ of the distance from P to Q, he meets A, who is on his return lap. What is the ratio of the speeds at which A and B run?

1. 3 : 2 2. 5 : 3 3. 5 : 4 4. 4 : 3

10. What is the distance between Debu's house and office, if he will reach office late by 20 minutes, traveling at 10 km/hr and will reach early by 15 minutes, traveling at 15 km/hr?

1. 14.5 km 2. 17.5 km 3. 16.5 km 4. 15 km

Answer key & Explanations

Q No.	Key	Explanation
1.	1	Circumference of path = πd = 22 / 7 × 161 = 506 yards. 3 mph = $\frac{22}{15}$ yards / sec. 5 mph = $\frac{22}{9}$ yards / sec. 6 mph = $\frac{44}{15}$ yards / sec. Time taken by *A*, *B* and *C* to complete the path = $\frac{506\times15}{22}=345$ sec, $\frac{506\times9}{22}=207$ sec and $\frac{506\times15}{44}=172.5$ sec. So they will be meeting again after LCM of 345, 207 & 172.5 i.e. after 1035 sec.
2.	3	Let the speed in still water = x km/hr. Takes 20 min. to row 12 km upstream ⇒ speed of u/s = 36 km/hr. Also time taken for u/s is $\frac{1}{3}$ more than for d/s. ∴ distance covered in d / s will be 1 / 3 more. Hence distance covered by man for d / s in 20 min. = 12 + $\frac{12}{3}$ = 16km. So speed of d / s = 48 km/hr. ∴ $x + y = 48$ and $x - y = 36$ ⇒ x = 42 km/hr.
3.	2	From the 2 relationships given, we can infer that *B* covers 50 – 10 = 40 m in 20 sec. ∴ *B* covers 50 m in 20 × $\frac{5}{4}=25$sec. ∴ *A* should give 25 sec. start to *B*, so that they finish together.
4.	3	A and B will meet at any point after A has taken one complete circle over B. Now the relative distance to be covered is 120 metres. Relative speed = 15 metres/sec. Time = 120/15 = 8 sec. Thus they will meet after 8 sec. A will reach the starting point after 120/40 sec. B will reach the starting point after 120/25 sec. You want them together, together means LCM i.e. = 120/5 = 24 sec. Now after 24 seconds they are meeting at the starting point, otherwise they are meeting after every 8 seconds. This means there will be 24/8 = 3 points where they will meet.
5.	1	Let the speed of the train be x mph and length of train = L miles. ∴ x = 180 L. Also, cycle is in opposite direction. ∴ Relative speed = x + 5. So $\frac{18}{60\times60}=\frac{L}{x+5}\Rightarrow 200L=x+5\Rightarrow 200L=180L+5\therefore L=\frac{1}{4}mile.$
6.	3	In the time taken by the boat in traveling 40 + (40 – 24) km, the eraser traveled 24 km. Let B be the speed of the boat in still water and W be the speed of the water. The two equations made will be $\frac{24}{W}=\frac{40}{S+W}+\frac{16}{S-W}$ (i) and $\frac{40}{S+W}+\frac{40}{S+W}=7$ (ii).

		Solving the two simultaneous equations get S = 14 and W = 6. Thus the downstream speed of the boat is 14 + 6 = 20 kmph.
7.	2	Let x be the length of train and y be the speed of train. $\therefore \frac{39}{60\times 60} = \frac{x}{y-5}$ and $\frac{33}{60\times 60} = \frac{x}{y+5}$. Solving these 2 equations, we get y = 60 m/hr.
8.	3	Let the required distance be $2x$. $\therefore$ Half i.e. x is covered at 45km/hr and other half at 55 km/hr. $\therefore \frac{x}{45} + \frac{x}{55} = 4 \Rightarrow x = 99$ $\therefore$ required distance = 99 × 2 = 198 km.
9.	2	Ratio of timing of A and B = $\frac{3}{4} : 1 + \frac{1}{4} = 3:5.$ Hence their speed ratio = 5:3.
10.	2	Difference in timing = 15 – (–20) = 35 min. Let required distance = x. $\therefore \frac{x}{10} - \frac{x}{15} = \frac{35}{60} \Rightarrow x = 17.5$ km.